B3.1 Gas exchange (Plant)

Notes written for this site. Slides, videos, worksheets and self-check questions: whatthecell.org/subtopics/sl-b3-1-plant.html

Written for this site from the course textbook and class slides. Use the page numbers to read the book's fuller account. Gas exchange in animals (B3.1.1–6) is in Body Systems. Image credits are on the last slide of the slides for this subtopic.

The leaf

B3.1.7 How leaves are adapted to exchange gases

pp. 283–286

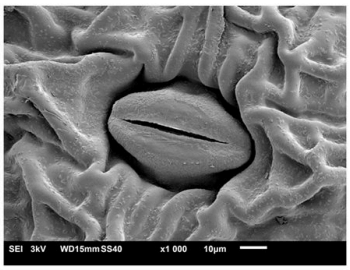
A leaf must take in carbon dioxide for photosynthesis, while limiting its loss of water. Being flat and thin gives it a large surface area and short diffusion distances.

Structure	Adaptation
Waxy cuticle	A largely impermeable layer, so gases and water vapour must pass through stomata, where flow is controlled
Epidermis	Most pores are in the lower epidermis, which is shaded and cooler, and this reduces water loss
Air spaces	Let gases circulate between loosely packed cells, maintaining gradients
Spongy mesophyll	A large, moist surface for gases to enter and leave cells
Stomatal guard cells	Open and close the pore, controlling gas exchange and water loss
Veins	Xylem supplies water, and phloem removes sugars, which helps to maintain gradients

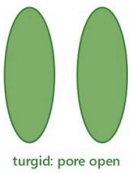
When guard cells take up water they become **turgid** and curve apart, opening the pore. When they lose water they become **flaccid** and the pore closes.

Guard cells control the stoma

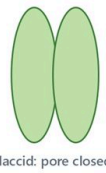
B3.1.7



Stoma and guard cells. Judyta Dulnik, CC BY-SA 4.0.



turgid: pore open



flaccid: pore closed

Water enters the guard cells, they become **turgid** and curve apart: the pore **opens**. When they lose water they become **flaccid** and it **closes**.

Common mistake
The stoma is the pore; guard cells are the two cells around it.

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Watch out: the stoma is the pore, and the guard cells are the cells around it.

B3.1.8 How tissues are arranged in a leaf

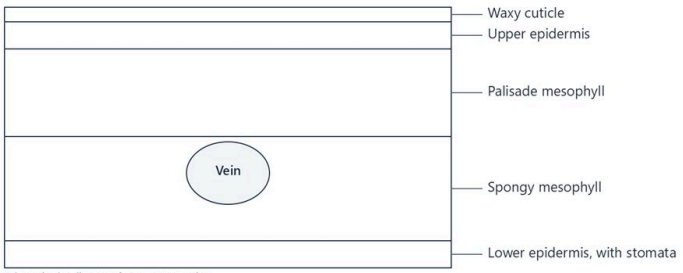
pp. 284–286

From top to bottom, a dicot leaf has a waxy cuticle, upper epidermis, palisade mesophyll, spongy mesophyll and lower epidermis with stomata. Veins, containing xylem and phloem, run through the mesophyll.

A **plan diagram** shows these tissues as regions. Use clear, continuous lines, do not shade, and do not draw individual cells.

Plan diagram of a dicot leaf

B3.1.8



Schematic plan diagram of a transverse section.

Plan diagram
Show tissues as regions. No individual cells, no shading.

Activity
Draw a plan diagram from a micrograph of a leaf section. Use continuous lines, keep the proportions, and label cuticle, epidermis, palisade and spongy mesophyll, xylem and phloem.

Losing water

B3.1.9 Why gas exchange in a leaf leads to transpiration

pp. 286–288

For carbon dioxide to enter, stomata must be open. Water evaporates from moist cell walls into the air spaces, and water vapour diffuses out through the same pores. This loss is **transpiration**.

Factor	Effect of an increase	Why
Temperature	Rate rises	More evaporation, so a greater concentration difference of water vapour
Humidity	Rate falls	The air outside holds more water vapour, so the gradient is smaller
Wind	Rate rises	Water vapour near the stomata is swept away, keeping the gradient steep
Light	Rate rises	Stomata open for photosynthesis, and the leaf warms

A **potometer** measures the rate at which a shoot takes up water, using the movement of an air bubble along a capillary tube. A small amount of this water stays in the plant, so it is an indirect measure.

Watch out: water vapour diffuses out, not liquid water, and this step is not osmosis.

B3.1.10 How many stomata a leaf has

pp. 289–291

Stomatal density is the number of stomata per unit area of leaf. More stomata can take up more carbon dioxide, but they can also lose more water.

1. Paint clear nail varnish on the underside of a leaf, and let it dry.
2. Lift it off with clear tape, and stick it to a slide.
3. Count the stomata in several fields of view, and find the mean.
4. Area of field = $3.14 \times r^2$. Density = mean count ÷ area.

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